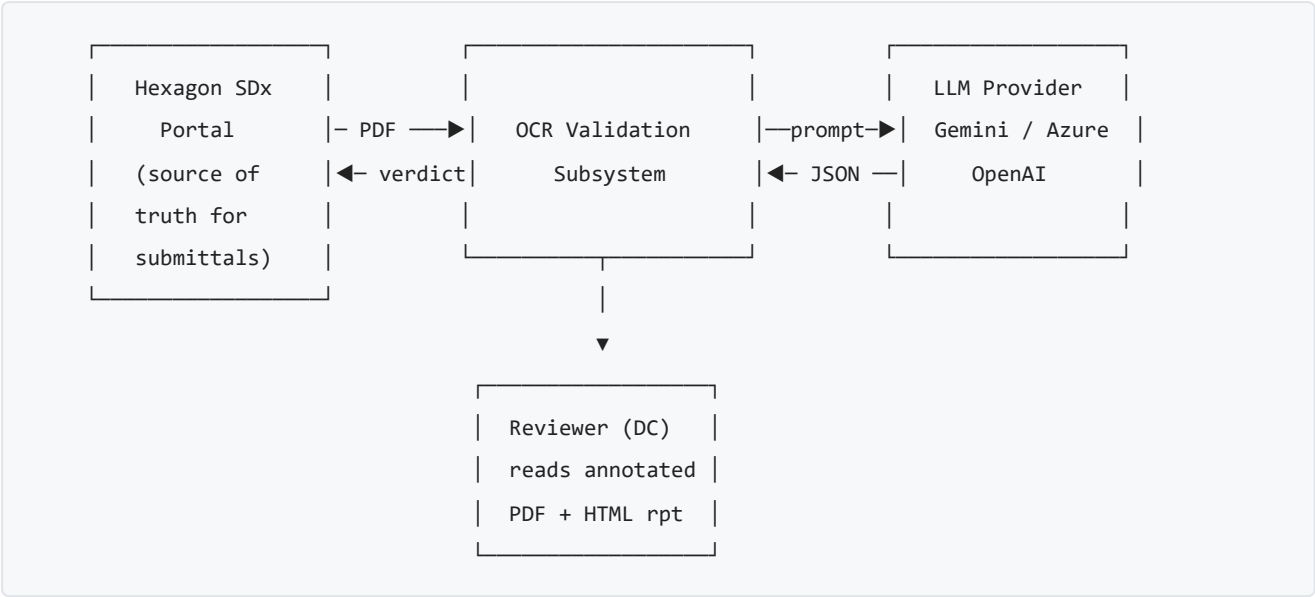


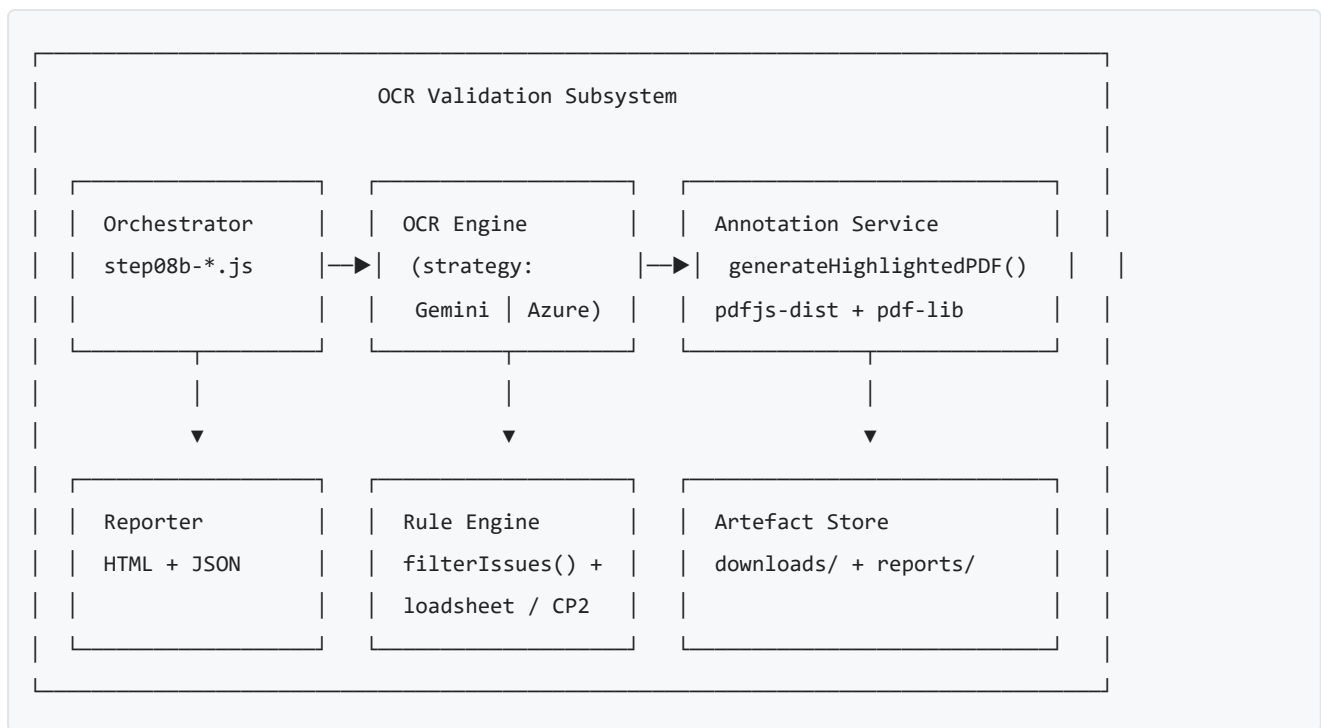
OCR Validation Subsystem — Architecture

Component: Document OCR Validation Pipeline **Location:** `hxgn-sdx-bot/bot/` — Step 8b of the SDx automation flow **Owner:** Document Control Automation **Date:** 2026-04-15

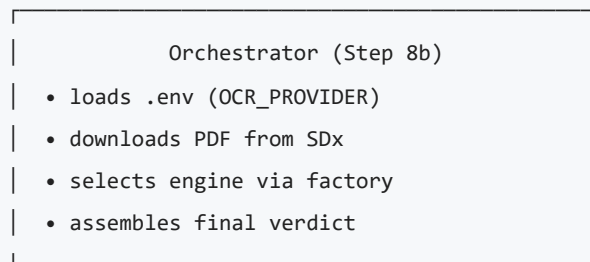
1. C4 — Context



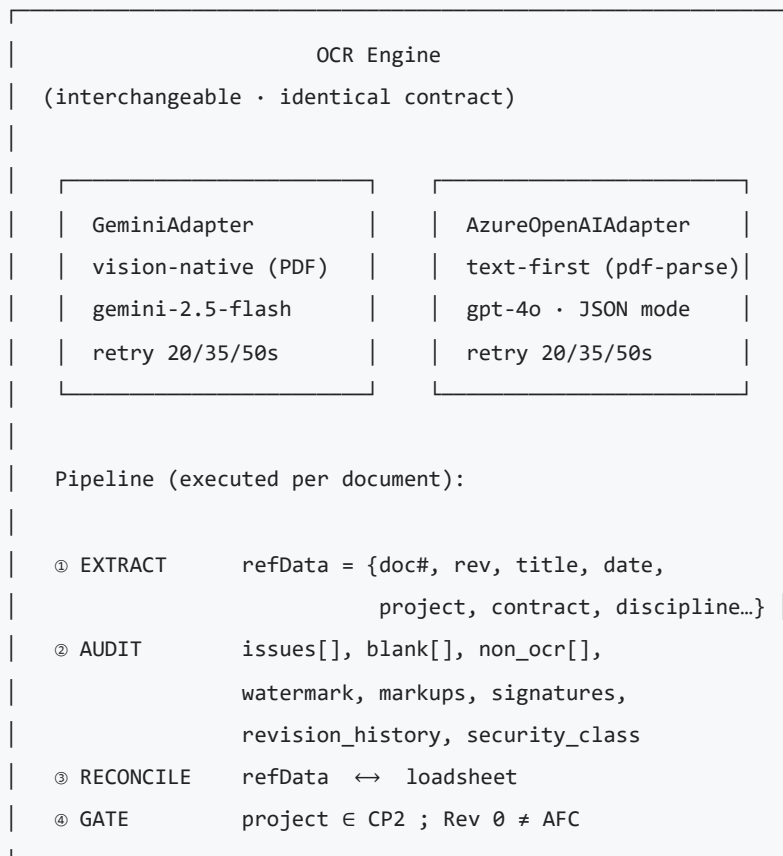
2. C4 — Containers



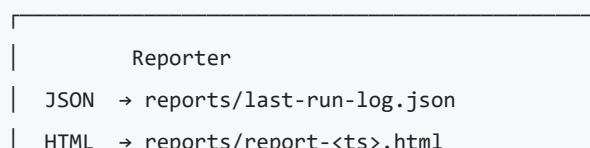
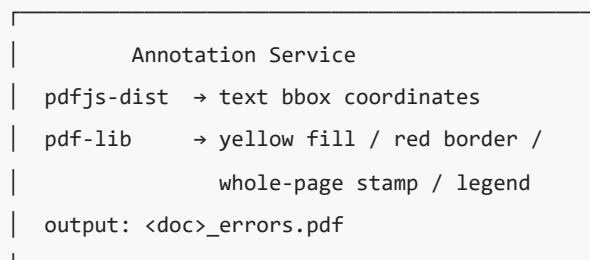
3. C4 — Components (runtime view)



runOCRValidation(pdf, loadsheet, submittal)

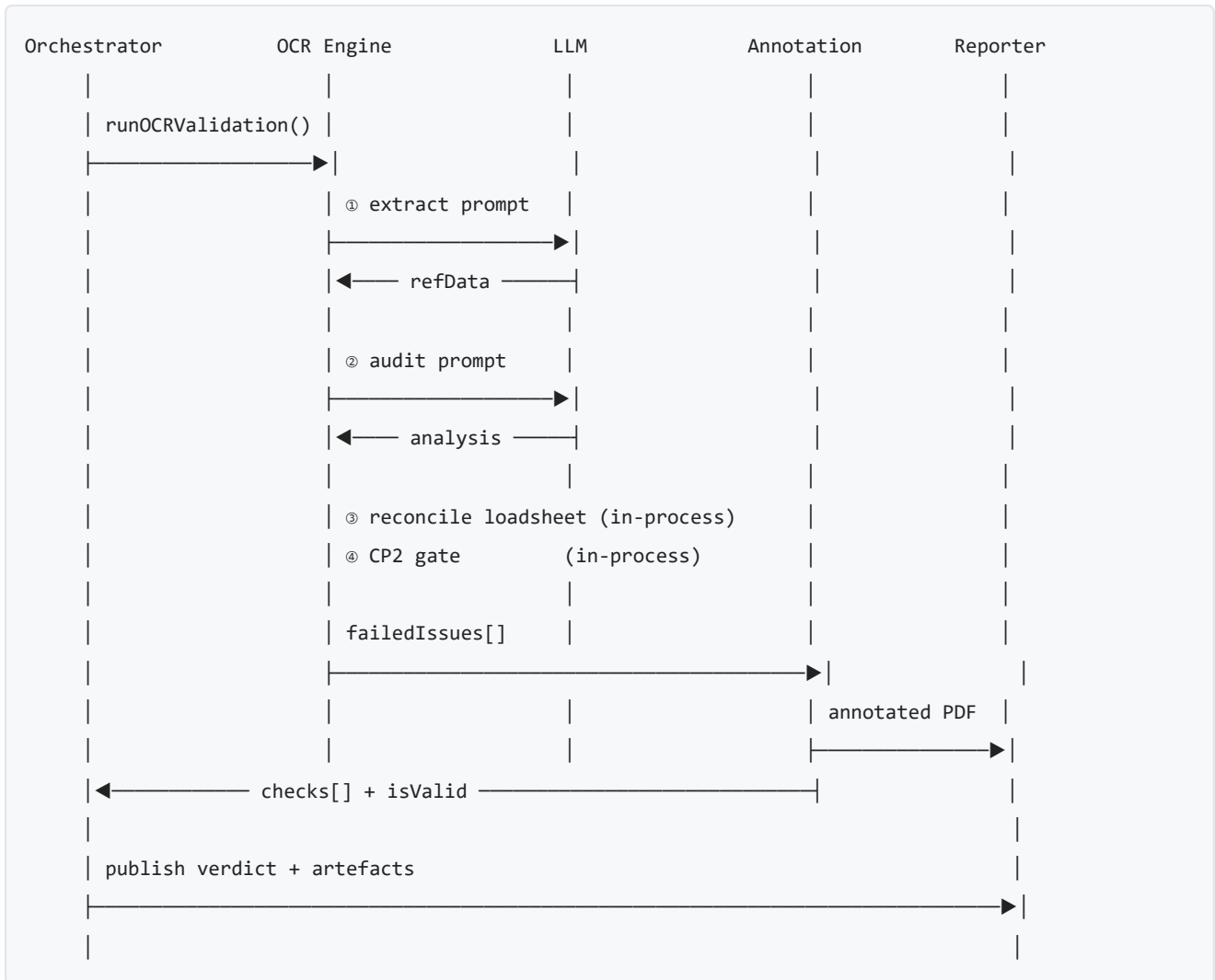


{ checks[], refData, analysis, isValid }



Verdict → SDx (ACCEPT / REJECT)

4. Sequence — single document validation



5. Design decisions & rationale

#	Decision	Rationale
1	Strategy pattern for LLM provider (<code>OCR_PROVIDER=gemini azure</code>)	Vendor independence; swap without touching the orchestrator. Identical public contract enforced by shared reducer.
2	Two-phase prompting (extract → audit)	Separates <i>ground truth capture</i> from <i>discrepancy hunting</i> ; keeps each prompt small and deterministic (<code>temperature=0</code> , JSON mode).
3	Deterministic override for non-OCR pages (Azure)	<code>pdf-parse</code> per-page char count is more reliable than the LLM for blank/image-only detection — a safety net against hallucination.
4	Post-LLM false-positive filter (<code>filterIssues</code>)	Normalises whitespace / case and reconciles 11 date formats before surfacing a failure — prevents reviewer fatigue.
5	Rule engine in code, not in the prompt (CP2 gate, loadsheet reconciliation)	Business rules are auditable, unit-testable, and cheap; only the linguistic work is delegated to the LLM.
6	Per-issue traceability ({ <code>page</code> , <code>field</code> , <code>found</code> , <code>expected</code> })	Enables precise bbox highlighting downstream; makes every failure defensible to an auditor.
7	Exponential back-off (20/35/50s) on 429	Matches both providers' published rate-limit recovery windows; avoids cascading failure during batch runs.

6. Quality attributes

Attribute	How it is achieved
Reliability	Retry with back-off; deterministic overrides; structured JSON responses.
Traceability	Every finding carries page + field + found + expected; artefacts persisted per run.
Explainability	Annotated PDF renders failures in-context for human reviewers.
Portability	Provider-agnostic interface; env-var switch; no prompt duplication beyond provider I/O shape.
Cost control	Text-first path (Azure) avoids vision tokens; Gemini path used only when native PDF is preferred.
Security	API keys in <code>.env</code> ; no PDF content logged; LLM calls use JSON-only response format.

7. Inputs / Outputs (contract)

Input

- `pdfPath` — absolute path to downloaded submittal PDF
- `loadsheets` — { `documentNumber`, `title`, `revision`, `discipline` }
- `submittal` — SDx metadata snapshot

Output

- `checks[]` — ordered list of {`name`, `status`: `PASS|WARN|FAIL`, `note`, `page?`, `field?`}
- `refData` — canonical metadata extracted from the PDF
- `analysis` — structural flags + filtered issues
- `isValid` — boolean gate consumed by Step 9 (approve/reject)
- `highlightedPdfPath` — annotated artefact for the reviewer

8. File map

Role	Path
Orchestrator	<code>hxgn-sdx-bot/bot/steps/step08b-document-validation.js</code>
Gemini adapter	<code>hxgn-sdx-bot/bot/utls/gemini-ocr.js</code>
Azure OpenAI adapter	<code>hxgn-sdx-bot/bot/utls/azure-openai-ocr.js</code>
Annotation service	<code>generateHighlightedPDF()</code> in <code>gemini-ocr.js</code>
Reporter	<code>hxgn-sdx-bot/bot/utls/reporter.js</code>
Artefact store	<code>hxgn-sdx-bot/bot/downloads/</code> · <code>hxgn-sdx-bot/bot/reports/</code>